

Chapter 2

Basic Physics of Semiconductors

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Basic Physics of Semiconductors

- Semiconductor Materials and Their Properties
- PN Junction
- Reverse Breakdown

Semiconductor Physics

▪ Semiconductor

- ✓ Semiconductor devices serve as heart of microelectronics.

▪ PN Junction

- ✓ PN junction is the most fundamental semiconductor device.

Semiconductors

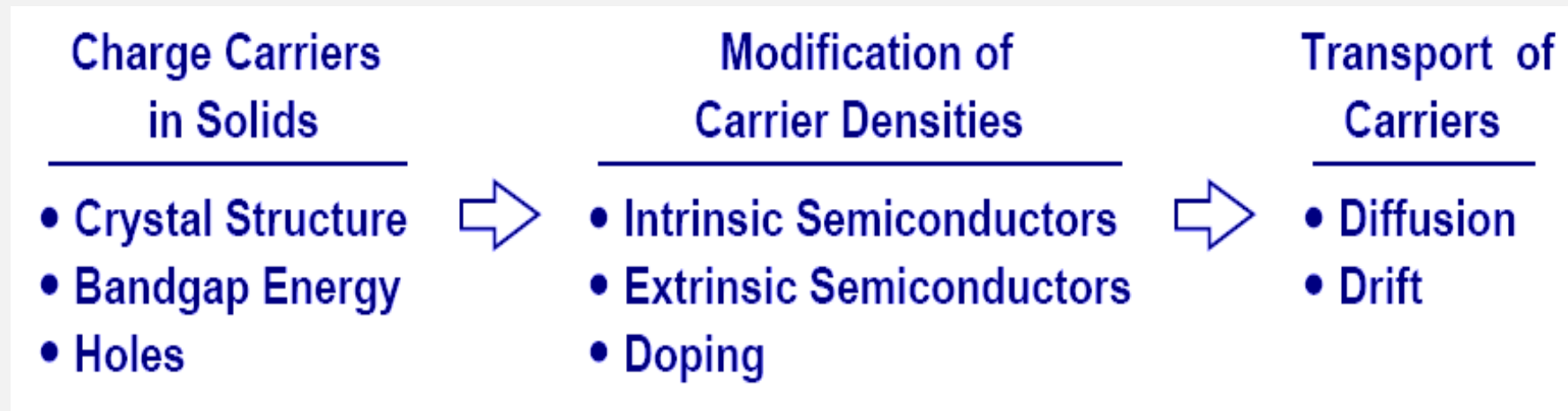
- Charge Carriers
- Doping
- Transport of Carriers

PN Junction

- Structure
- Reverse and Forward Bias Conditions
- I/V Characteristics
- Circuit Models

Semiconductor Materials and Their Properties

- To understand PN junction's **IV characteristics**, it is important to understand charge carriers' **behavior** in solids, how to modify carrier **densities**, and different **mechanisms** of charge flow.



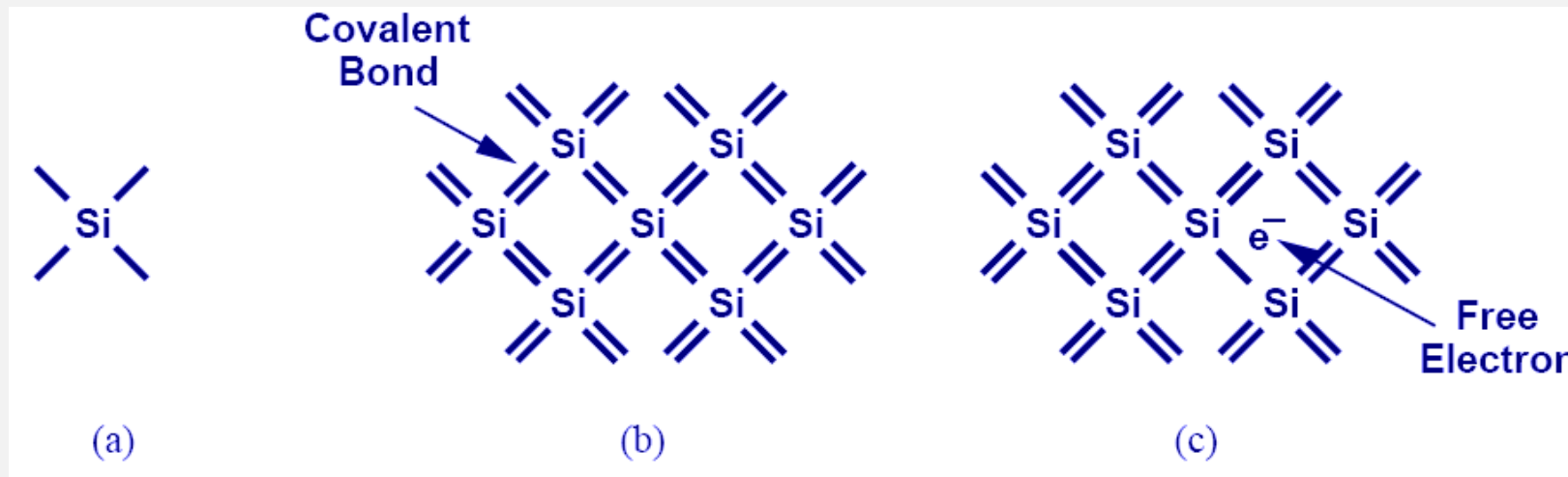
Periodic Table

- This abridged table contains elements with **3 to 5 valence** electrons, with Si being the most important.

III		IV	V		
		Boron (B)	Carbon (C)		
• • •	Aluminum (Al)	Silicon (Si)	Phosphorous (P)	• • •	
	Galium (Al)	Germanium (Ge)	Arsenic (As)		
		• • •			

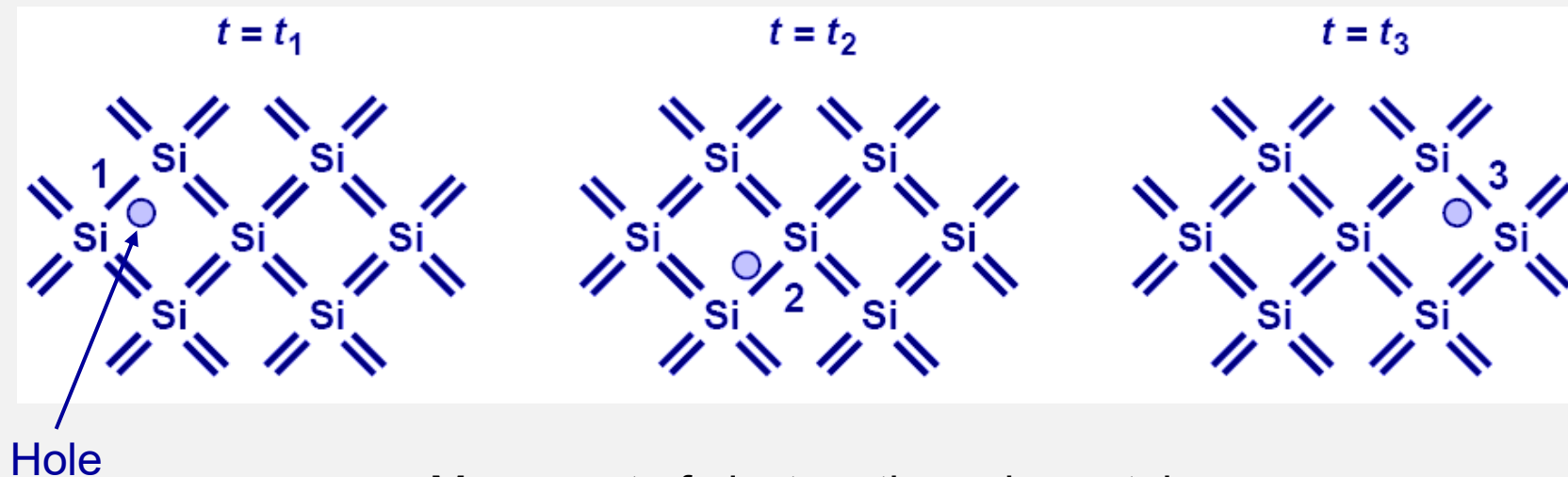
Silicon

- Si has four **valence** electrons. Therefore, it can form **covalent** bonds with four of its neighbors.
- When **temperature** goes up, electrons in the covalent bond can become **free**.



Electron-Hole Pair Interaction

- With free electrons **breaking off** covalent bonds, holes are generated.
- Holes can be filled by **absorbing** other free electrons, so effectively there is a flow of charge carriers.



Movement of electron through crystal

Free Electron Density at a Given Temperature

- E_g , or **bandgap energy** determines how much effort is needed to break off an electron from its covalent bond.
- There exists an **exponential** relationship between the free-electron density and bandgap energy.

$$n_i = 5.2 \times 10^{15} T^{3/2} \exp \frac{-E_g}{2kT} \text{ electrons/cm}^3$$

where $k = 1.38 \times 10^{-23} \text{ J/K}$ is called the Boltzmann constant.

$$\checkmark E_g \uparrow \Rightarrow n_i \downarrow$$

$$\checkmark T \rightarrow 0, n_i \rightarrow 0$$

E_g for Different Materials

- **Insulators** display a **high** E_g ; for example, $E_g = 2.5 \text{ eV}$ for diamond.
- **Conductors**, on the other hand, have a **small** bandgap.
- **Semiconductors** exhibit a **moderate** E_g , typically ranging from **1 eV to 1.5 eV**.
- The unit **eV (electron volt)** represents the energy necessary to move one electron across a potential difference of 1V.
($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$)

Example 2-1

- Determine the density of electrons in **silicon** at $T = 300$ K (room temperature) and $T = 600$ K.

Solution:

Since $E_g = 1.12 \text{ eV} = 1.792 \times 10^{-19} \text{ J}$, we have

$$n_i = 5.2 \times 10^{15} T^{3/2} \exp \frac{-E_g}{2kT} \text{ electrons/cm}^3$$

$$n_i(T = 300 \text{ K}) = 1.08 \times 10^{10} \text{ electrons/cm}^3$$

$$n_i(T = 600 \text{ K}) = 1.54 \times 10^{15} \text{ electrons/cm}^3$$

- Silicon has $5 \times 10^{22} \text{ atoms/cm}^3$, only $\sim \frac{1}{5 \times 10^{12}} \text{ electron/atoms}$ at room temperature.
- In other words, **silicon** still seems a **very poor conductor**.

Exercise

- Repeat the above exercise for a material having a bandgap of **1.5 eV**.

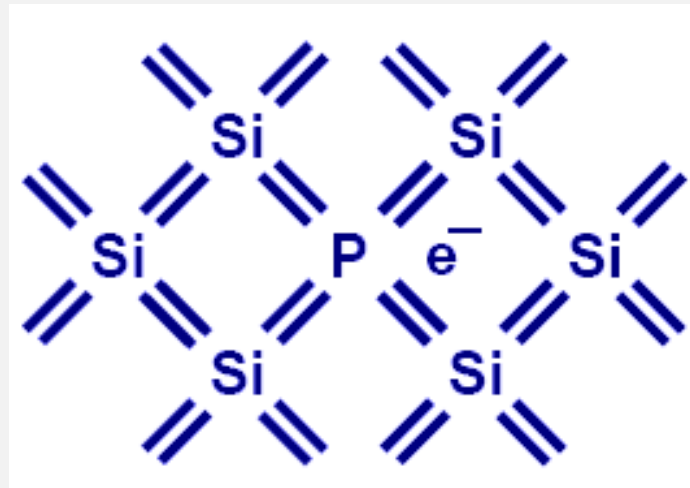
Modification of Carrier Densities

Intrinsic vs. Extrinsic Semiconductors

- **Problem:** Intrinsic (“pure” type) silicon suffering from a **very high resistance**.
- Fortunately, it is possible to modify the resistivity of silicon by replacing some of the atoms in the crystal with atoms of another material.
 - ✓ **Doping: Intrinsic $\Rightarrow$ Extrinsic ($\downarrow$ resistance)**

Doping (N type)

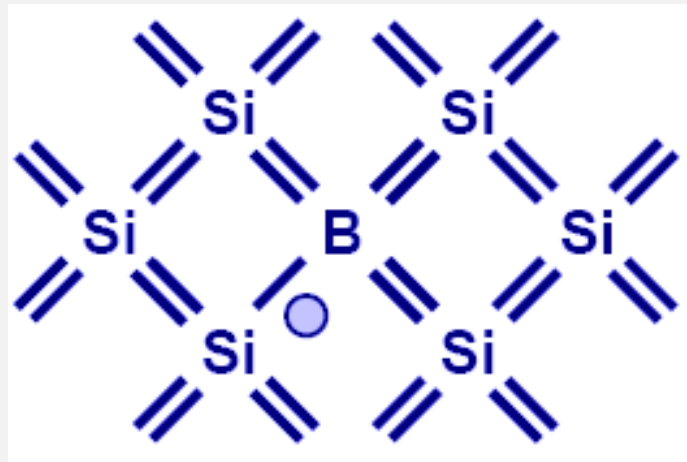
- Pure Si can be doped with other elements (“**dopant**”) to change its electrical properties.
- For example, if **Si** is doped with **P** (**phosphorous**), then it has more electrons, or becomes type **N** (**electron**). Thus, P is named as “**donor**”.



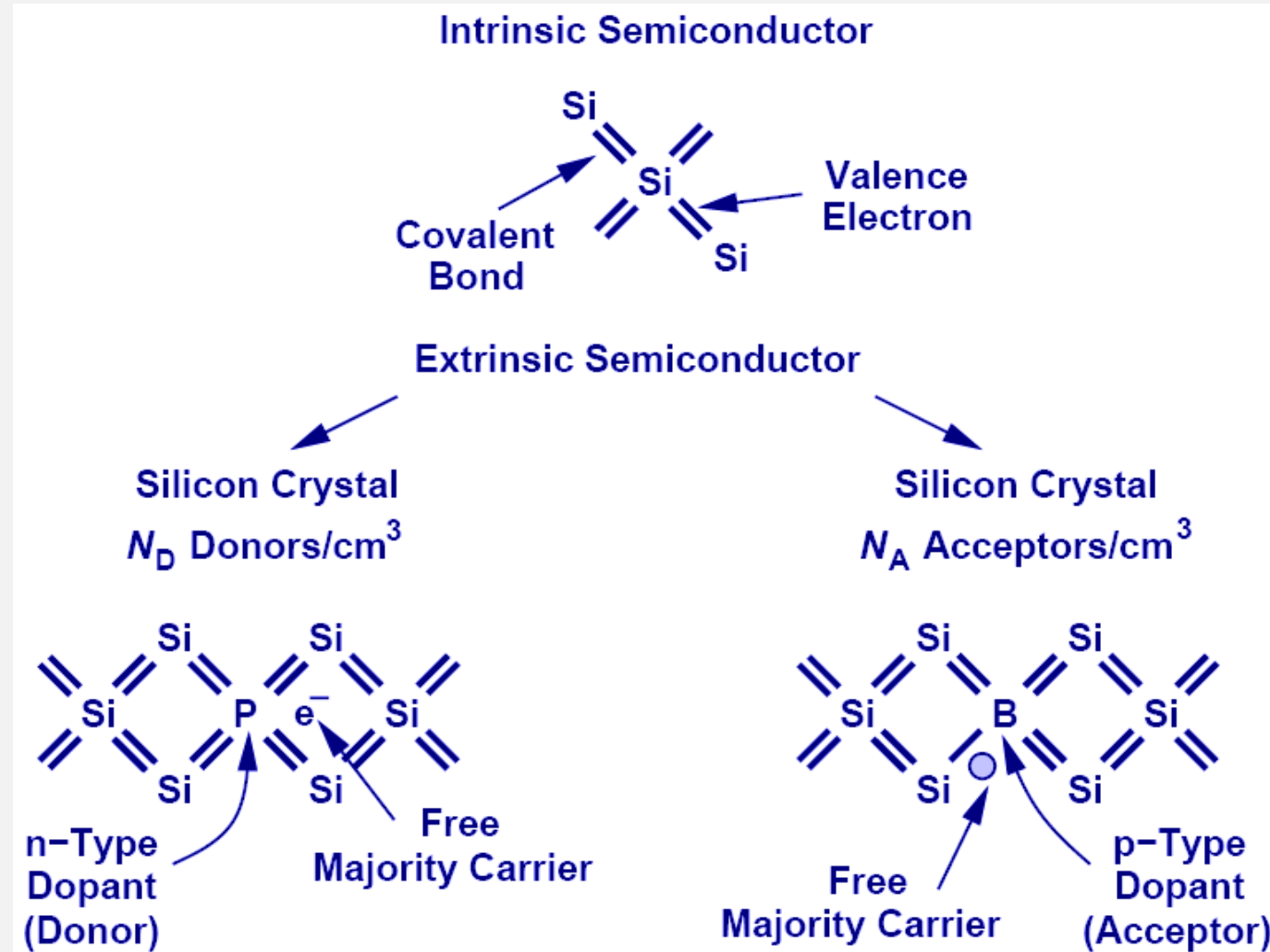
Loosely-attached electron with phosphorus doping

Doping (P type)

- If **Si** is doped with **B** (boron), then it has more holes, or becomes type **P** (hole). Thus, B is named as “**acceptor**”.



Summary of Charge Carriers



Electron and Hole Densities

- In an intrinsic semiconductor, the electron density, $n(=n_i)$ is equal to the hole density, p .

$$np = n_i^2$$

- The product of electron and hole densities is **ALWAYS** equal to the square of intrinsic electron density **regardless of doping levels**.
- For an extrinsic semiconductor,

$$np = n_i^2 \quad (\text{Mass Action Law})$$

Electron and Hole Densities

- For extrinsic semiconductors,

- ✓ N type:

Majority Carriers: $n \approx N_D$ ($\because N_D \gg n_i$)

Minority Carriers: $p \approx \frac{n_i^2}{N_D}$

- ✓ P type:

Majority Carriers: $p \approx N_A$ ($\because N_A \gg n_i$)

Minority Carriers: $n \approx \frac{n_i^2}{N_A}$

Note: Either n-type or p-type silicon is electrically neutral.

Example 2-3

■ A piece of crystalline silicon is doped uniformly with phosphorus atoms. The doping density is 10^{16} electrons/cm³. Determine the electron and hole densities in this material at the room temperature.

Solution:

Since the free electrons donated by phosphorus exceeds the free electrons of Si by $10^6 = 10^{16}/10^{10}$ (i.e., $N_D \gg n_i$), we can assume

$$n \approx N_D = 10^{16} \text{ electrons/cm}^3$$

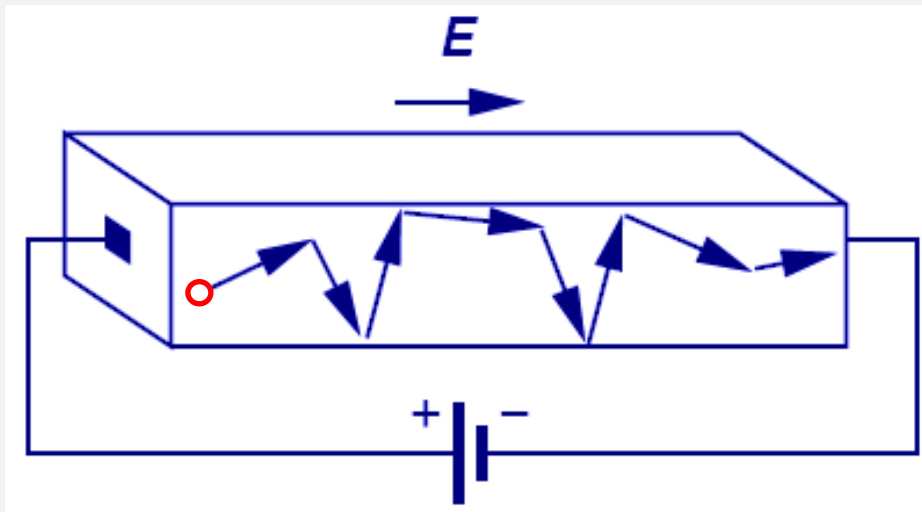
$$\because n_i (T = 300 \text{ K}) = 1.08 \times 10^{10} \text{ electrons/cm}^3$$

$$p \approx \frac{n_i^2}{N_D} = \frac{(1.08 \times 10^{10})^2}{10^{16}} = 1.17 \times 10^4 \text{ holes/cm}^3$$

➤ This example justifies the reason for calling electrons the “majority carriers” and holes the “minority carriers” in an n-type semiconductor.

Drift (First Charge Transportation Mechanism)

- The process in which charge particles move because of an electric field is called drift.
- Charge particles will move at a velocity (v) that is proportional to the electric field, i.e., $v = \mu E$, where μ is the mobility.



$$\begin{aligned}\vec{v}_h &= \mu_p \vec{E} \\ \vec{v}_e &= -\mu_n \vec{E}\end{aligned}$$

Example 2-5

- A uniform piece of n-type of silicon that is $1\text{ }\mu\text{m}$ long senses a voltage of 1 V. Determine the velocity of the electrons.

Solution:

$$E = \frac{V}{L} = \frac{1}{10^{-6}} = 10^6 \text{ V/m}$$

$\therefore$ the mobility of electrons, $\mu_n = 1350 \text{ cm}^2/(\text{V} \cdot \text{s})$

$$v_e = \mu_n E = 1.35 \times 10^7 \text{ cm/s}$$

➤ In other words, electrons take

$$\Delta t = L/v_e = (1\text{ }\mu\text{m})/(1.35 \times 10^7 \text{ cm/s}) = 7.4 \text{ ps}$$

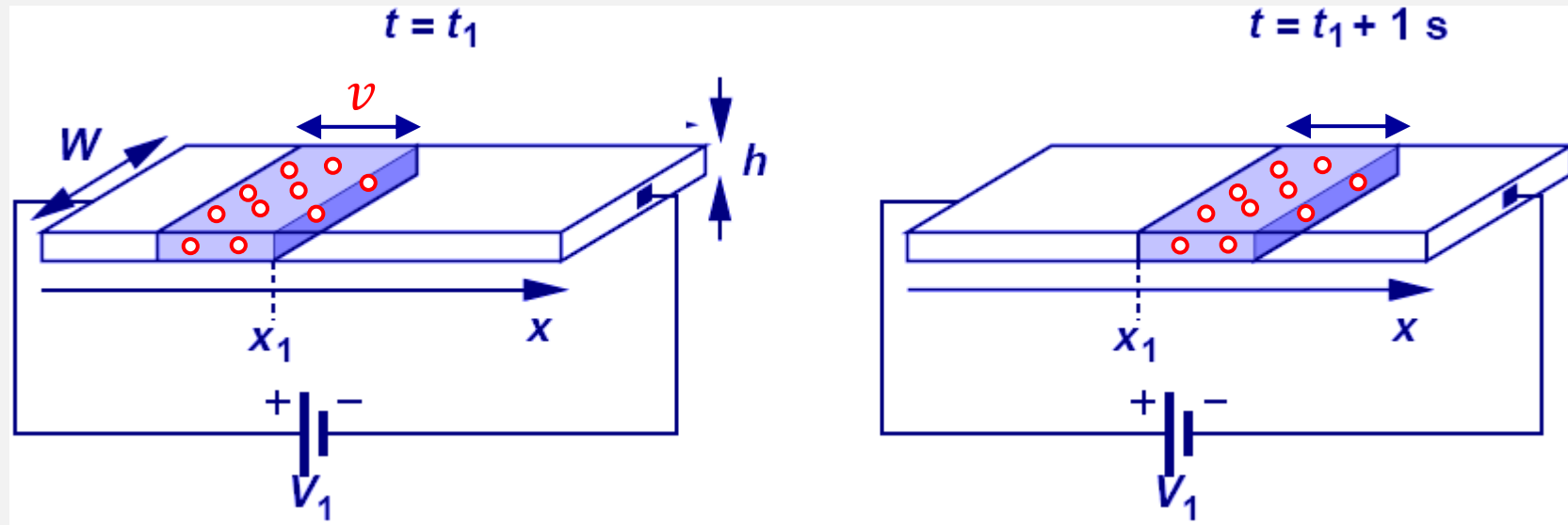
to cross the $1\text{ }\mu\text{m}$ length.

Current Flow of Drift

- Electric current I_p is calculated as the amount of charge in v meters that passes through a cross-section if the charge travel with a velocity of v m/s.

$$I_p = (v \cdot W \cdot h) \cdot p \cdot q$$

where $q = 1.6 \times 10^{-19}$ C is an electron charge.



Total Drift Current Density

- The hole drift current density (J_p): the current passing through a unit cross section.

$$\because J_p = \frac{I_p}{A} = \frac{(v \cdot W \cdot h) \cdot p \cdot q}{Wh} = v \cdot p \cdot q$$
$$J_p = \mu_p E \cdot p \cdot q \text{ (A/cm}^2\text{)}$$

- Similarly, the electron drift current density (J_n) is

$$J_n = \mu_n E \cdot n \cdot q \text{ (A/cm}^2\text{)}$$

- The total drift current density (J_{tot}): the drift current density in a semiconductor includes electron and hole drift densities.

$$J_{tot} = J_n + J_p = \mu_n E \cdot n \cdot q + \mu_p E \cdot p \cdot q$$
$$= q(\mu_n n + \mu_p p)E$$

Example 2-6

- In an experiment, it is desired to obtain equal electron and hole drift currents. How should the carrier densities be chosen?

Solution:

$$\because I_n = I_p \Rightarrow J_n = J_p \Rightarrow \mu_n n = \mu_p p$$

$$\therefore \frac{n}{p} = \frac{\mu_p}{\mu_n} \quad (1)$$

$$\because np = n_i^2 \quad (2)$$

$$\text{From (1), } n = \frac{\mu_p}{\mu_n} p$$

$$\left(\frac{\mu_p}{\mu_n} p\right)p = n_i^2 \Rightarrow p^2 = \frac{\mu_n}{\mu_p} n_i^2$$

$$p = \sqrt{\frac{\mu_n}{\mu_p}} n_i$$

$$\text{Similarly, } n = \sqrt{\frac{\mu_p}{\mu_n}} n_i$$

$$\text{For silicon, } \frac{\mu_n}{\mu_p} = \frac{1350}{480} = 2.81$$

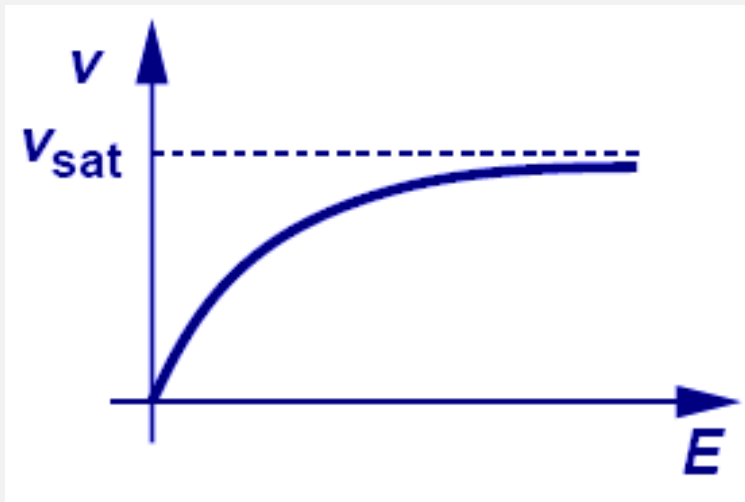
$$p = 1.68 n_i$$

$$n = 0.596 n_i$$

- Since p and n are of the same order as n_i equal electron and hole drift currents can occur for only a very lightly doped material.

Velocity Saturation

- So far, we assume the the **mobility** of carriers in semiconductors is *independent* of the **electric field**, and the carrier velocity increases **linearly** with E according to $v = \mu E$.
- **Velocity saturation**: In reality, carrier velocity does **not increase linearly** with the electric field due to **frequent collisions with lattice** and eventually **saturates** at high electric fields, **limiting transistor performance**.

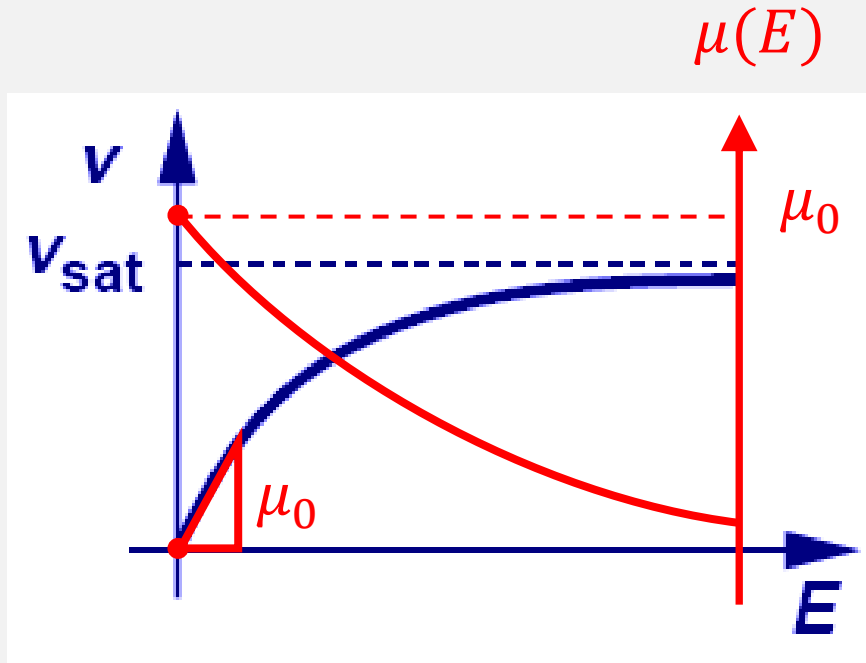


$$v(E) = \left(\frac{\mu_0}{1 + \frac{\mu_0 E}{v_{sat}}} \right) E$$

where μ_0 is the “low-field” mobility.

“Effective” Mobility: $\mu(E)$

- The “effective” mobility $\mu(E)$ is given by the following expression and gradually decreases toward zero as E increases



$$\mu(E) = \frac{\mu_0}{1 + \frac{\mu_0}{v_{sat}} E}$$

- The carrier velocity is given by $v(E) = \mu(E) \cdot E = \left(\frac{\mu_0}{1 + \mu_0 E / v_{sat}} \right) E$.

Example 2-7

- A uniform piece of semiconductor $0.2 \mu\text{m}$ long sustains a voltage of 1 V . If the low-field mobility is equal to $1350 \text{ cm}^2/(\text{V} \cdot \text{s})$ and the saturation velocity of the carriers 10^7 cm/s , determine the effective mobility. Also, calculate the maximum allowable voltage such that the effective mobility is only 10% lower than μ_0 .

Solution:

$$\begin{aligned} E &= \frac{V}{L} = \frac{1}{0.2 \times 10^{-6}} \\ &= 5 \times 10^6 \text{ V/m} = 5 \times 10^4 \text{ V/cm} \\ \therefore \mu_0 &= 1350 \text{ cm}^2/(\text{V} \cdot \text{s}) \\ \mu &= \frac{\mu_0}{1 + \frac{\mu_0 E}{v_{sat}}} = \frac{1350}{1 + \frac{(1350) \times (5 \times 10^4)}{10^7}} \\ &= \frac{\mu_0}{7.75} = 0.129 \mu_0 = 174 \text{ cm}^2/(\text{V} \cdot \text{s}) \end{aligned}$$

For $\mu = 0.9 \mu_0$,

$$\begin{aligned} \therefore 0.9 \mu_0 &= \frac{\mu_0}{1 + \frac{\mu_0 E}{v_{sat}}} \\ E &= \frac{v_{sat}}{9 \mu_0} = 823 \text{ V/cm} \\ V &= EL = 823 \times 0.2 \times 10^{-6} \\ &= 16.5 \text{ mV} \end{aligned}$$

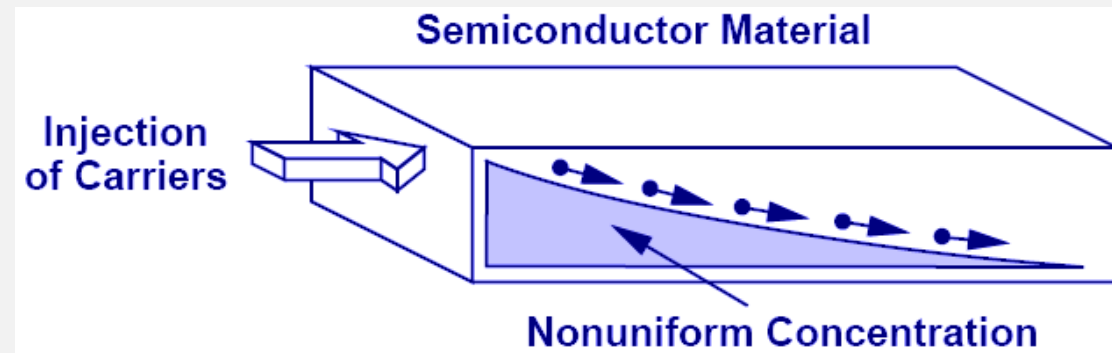
- This example shows that modern submicron devices experience velocity saturation because they operate at voltages much higher than 16.5 mV .

Exercise

- At what voltage does the mobility fall by 20%?

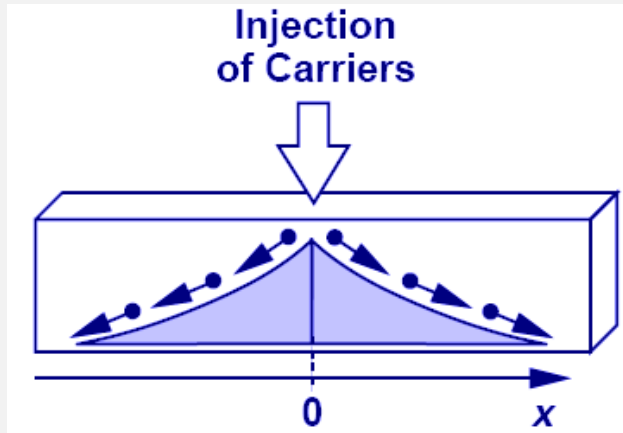
Diffusion (Second Charge Transportation Mechanism)

- Charge particles move from a region of high concentration to a region of low concentration. It is analogous to an every day example of an ink droplet in water.



Diffusion Current

- Diffusion current is proportional to the **gradient of charge** ($\frac{dn}{dx}$) along the direction of current flow.



$$I_n = AqD_n \frac{dn}{dx}$$

where D_n is a proportionality factor called the “diffusion constant” .

- Its total current density consists of both electrons and holes.

$$J_n = qD_n \frac{dn}{dx} , J_p = -qD_p \frac{dp}{dx}$$

$$J_{tot} = q(D_n \frac{dn}{dx} - D_p \frac{dp}{dx})$$

For intrinsic silicon,

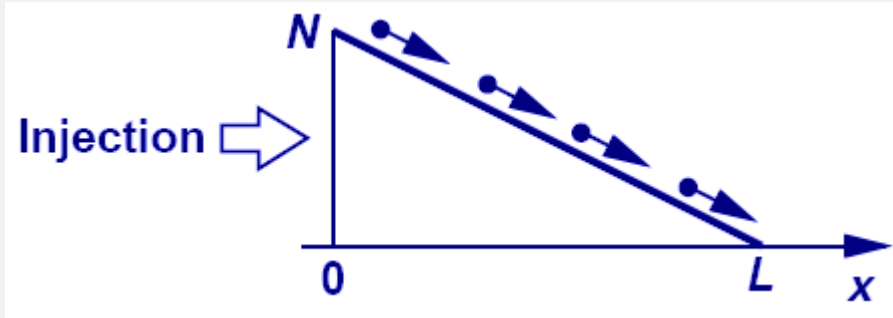
$$D_n = 34 \text{ cm}^2/\text{s} \text{ (for electrons)}$$

$$D_p = 12 \text{ cm}^2/\text{s} \text{ (for holes)}$$

Example 2-8

- Suppose the electron concentration is equal to N at $x = 0$ and falls linearly to zero at $x = L$. Determine the diffusion current.

Solution:



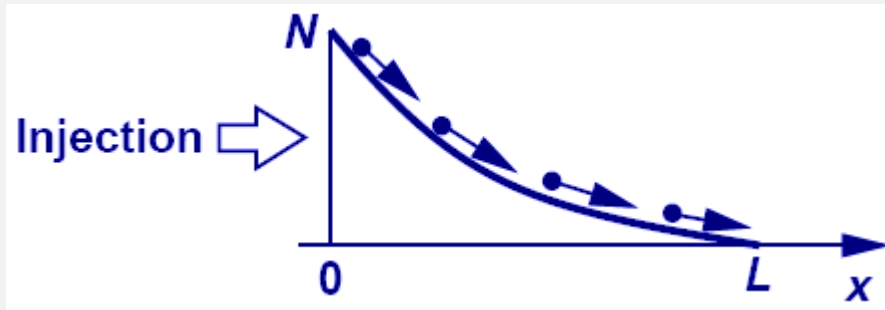
$$\begin{aligned} J_n &= qD_n \frac{dn}{dx} \\ &= -qD_n \cdot \frac{N}{L} \end{aligned}$$

➤ The current is constant along the x-axis.

Example 2-9

- Repeat the above example but assume an exponential gradient, $n(x) = N \exp \frac{-x}{L_d}$, where L_d is a constant.

Solution:

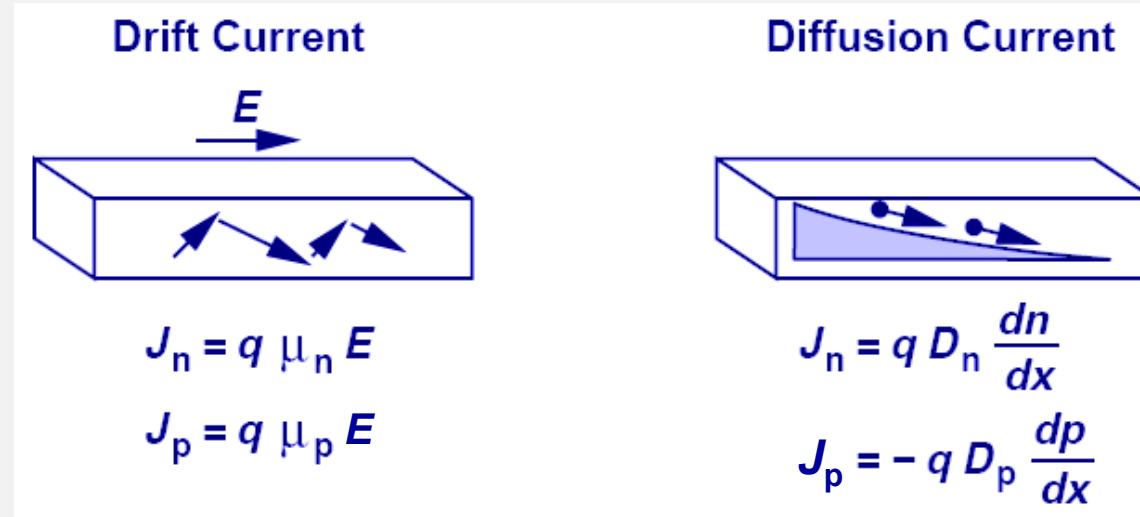


$$\begin{aligned} J_n &= qD_n \frac{dn}{dx} \\ &= \frac{-qD_n N}{L_d} \exp \frac{-x}{L_d} \end{aligned}$$

➤ The current is constant along the x-axis.

Einstein's Relation

- While the underlying physics behind drift and diffusion currents are totally different.



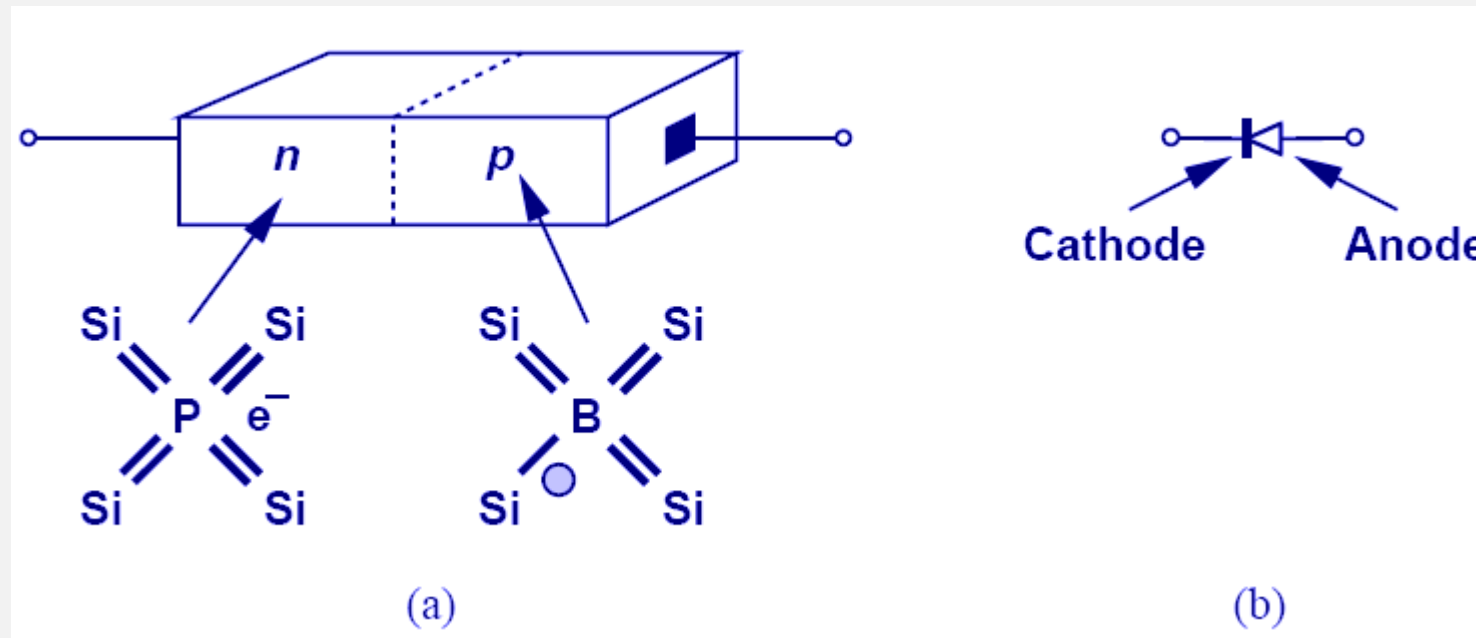
- Einstein's relation** provides a mysterious link between the two, i.e., μ_n (or μ_p) and D_n (or D_p).

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q}$$

- ✓ Note that $V_T \triangleq \frac{kT}{q}$ is known as “thermal voltage”.
- ✓ $V_T \approx 26 \text{ mV}$ at $T = 300 \text{ K}$.

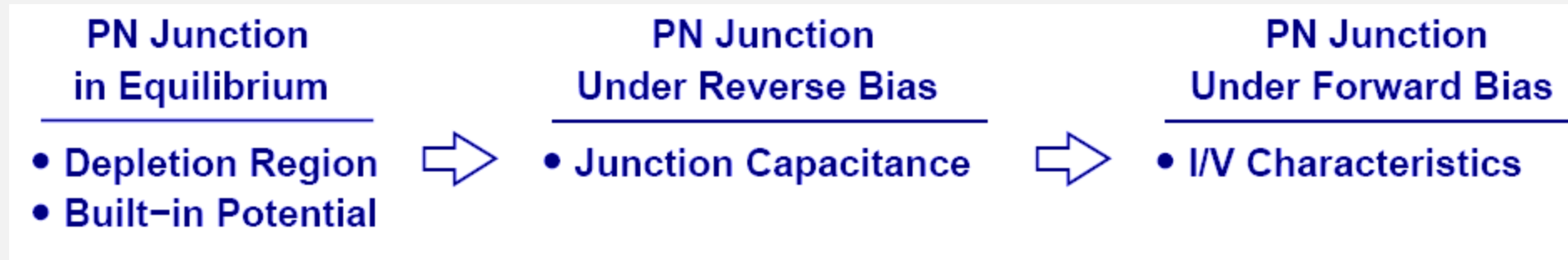
PN Junction (Diode)

- When N-type and P-type dopants are introduced side-by-side in a semiconductor, a PN junction or a diode is formed.



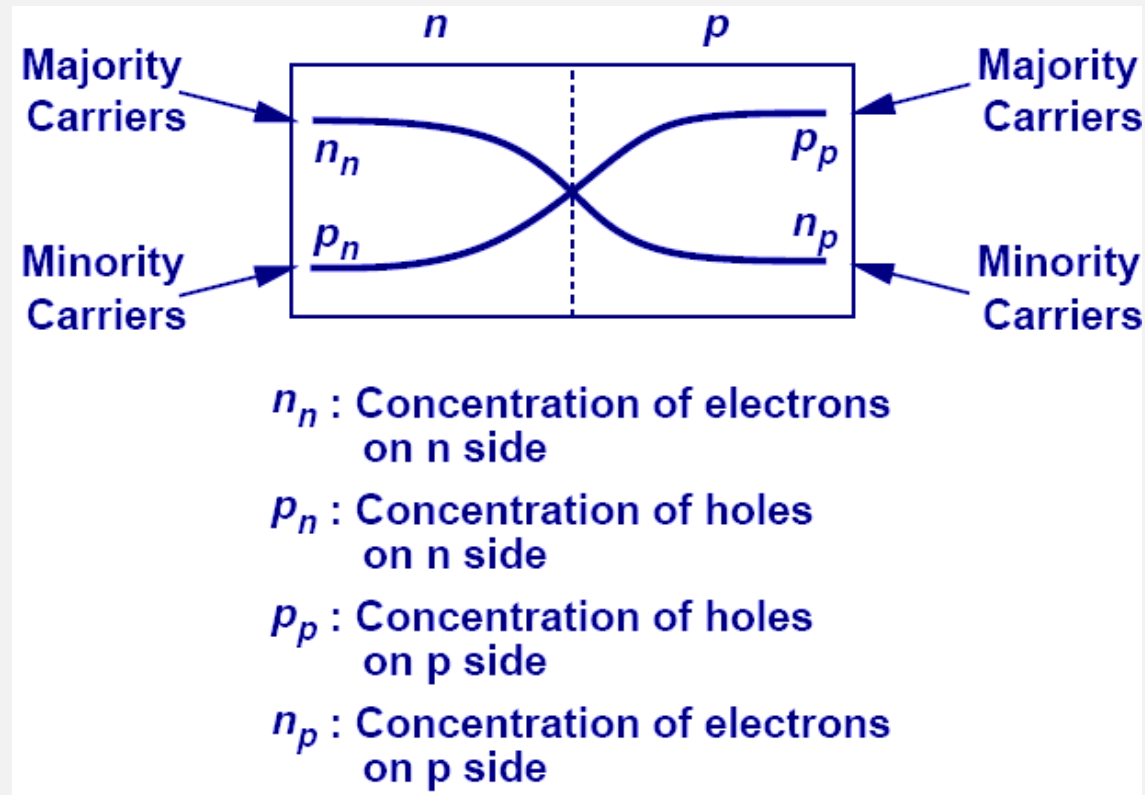
Diode's Three Operation Regions

- In order to understand the operation of a diode, it is necessary to study its three operation regions: equilibrium, reverse bias, and forward bias.



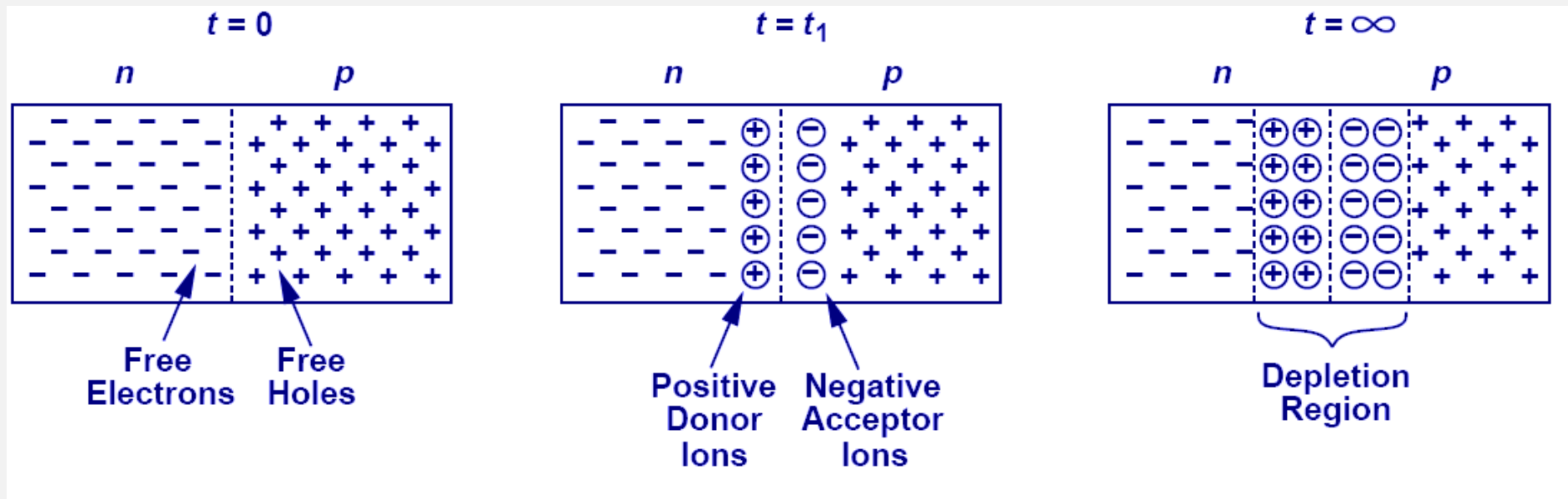
pn Junction in Equilibrium

- Because each side of the junction contains an excess of holes or electrons compared to the other side, there exists a large concentration gradient. Therefore, a **diffusion** current flows **across the junction** from each side.



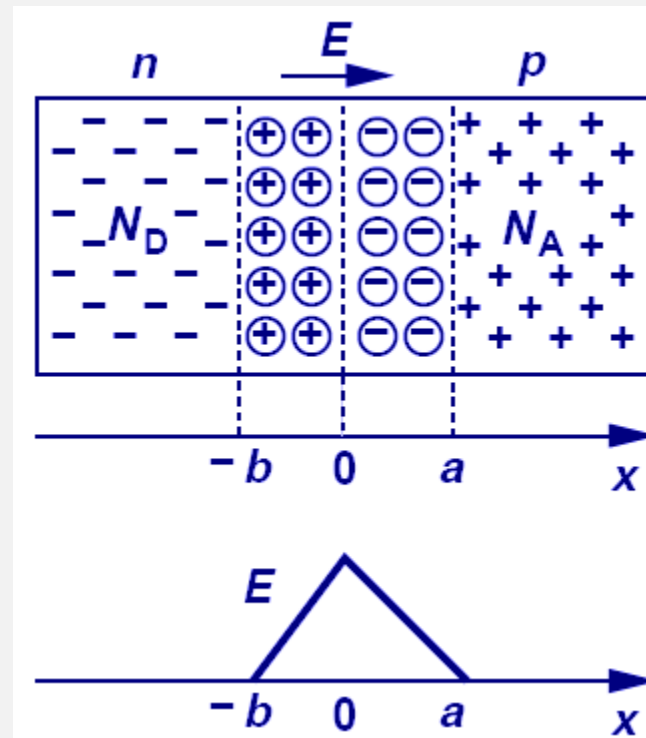
Depletion Region

- As free electrons and holes diffuse across the junction, a region of **fixed ions** is left behind. This region is known as the “**depletion region**”.



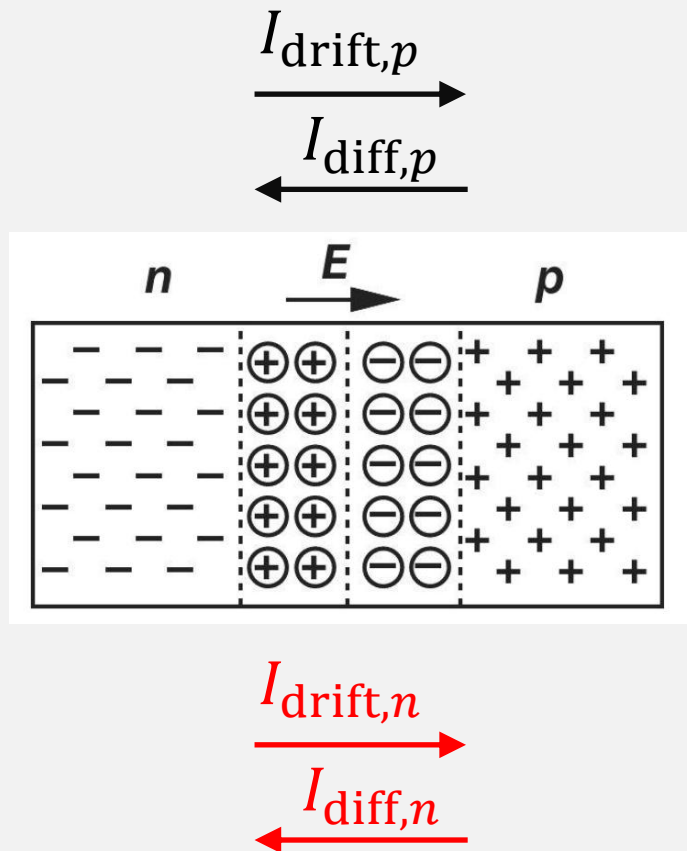
Drift Current

- The fixed ions in depletion region create an electric field that results in a **drift** current.



Equilibrium

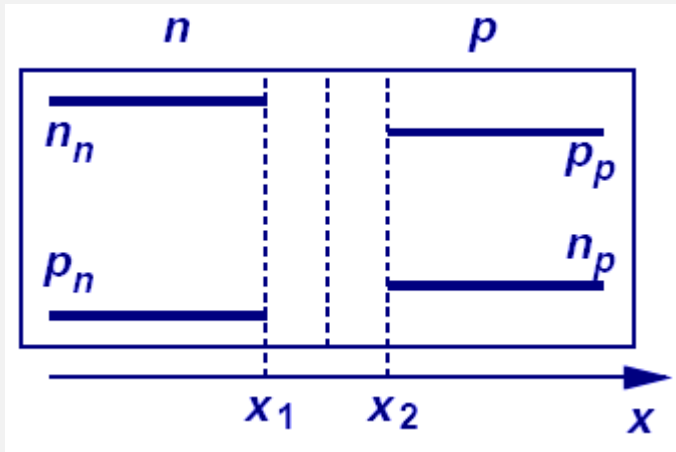
- Once **the electric field is strong enough** to completely stop the diffusion currents.
- In other words, the **drift** currents resulting from the **electric field** exactly cancel the **diffusion** currents due to the **gradients**.



$$I_{\text{drift},p} + I_{\text{diff},p} = 0$$
$$I_{\text{drift},n} + I_{\text{diff},n} = 0$$

Built-in Potential

- Because of the electric field across the junction, there exists a built-in potential. Its derivation is shown below.



Carrier profiles in a pn junction

$$\because I_{\text{drift},p} + I_{\text{diff},p} = 0 \Rightarrow \because J_{\text{drift},p} = -J_{\text{diff},p}$$

$$q\mu_p p E = qD_p \frac{dp}{dx}$$

$$\because E = -\frac{dV}{dx} \Rightarrow -\mu_p p \frac{dV}{dx} = D_p \frac{dp}{dx}$$

$$-\mu_p dV = D_p \frac{dp}{p}$$

$$-\mu_p \int_{x_1}^{x_2} dV = D_p \int_{p_n}^{p_p} \frac{dp}{p}$$

$$V(x_1) - V(x_2) = \frac{D_p}{\mu_p} \ln \frac{p_p}{p_n}$$

From Einstein's relation,

$$\frac{D_p}{\mu_p} = \frac{kT}{q}$$

$$\begin{aligned}\therefore V(x_1) - V(x_2) &= \frac{D_p}{\mu_p} \ln \frac{p_p}{p_n} \\ &= \frac{kT}{q} \ln \frac{p_p}{p_n}\end{aligned}$$

$$V_0 \triangleq V(x_1) - V(x_2)$$

$$\because p_p = N_A, \text{ and } p_n = \frac{n_i^2}{N_D}$$

$$V_0 = \frac{kT}{q} \ln \frac{p_p}{p_n} = V_T \ln \frac{N_A N_D}{n_i^2}$$

Example 2-13

- A silicon pn junction employs $N_A = 2 \times 10^{16} \text{ cm}^{-3}$ and $N_D = 4 \times 10^{16} \text{ cm}^{-3}$. Determine the built-in potential at room temperature ($T = 300 \text{ K}$).

Solution:

From Example 2-1, $n_i(T = 300 \text{ K}) = 1.08 \times 10^{10} \text{ electrons/cm}^3$.

Thus,

$$\begin{aligned} V_0 &= (26 \text{ mV}) \ln \frac{(2 \times 10^{16}) \times (4 \times 10^{16})}{(1.08 \times 10^{10})^2} \\ &= 768 \text{ mV} \end{aligned}$$

Example 2-14

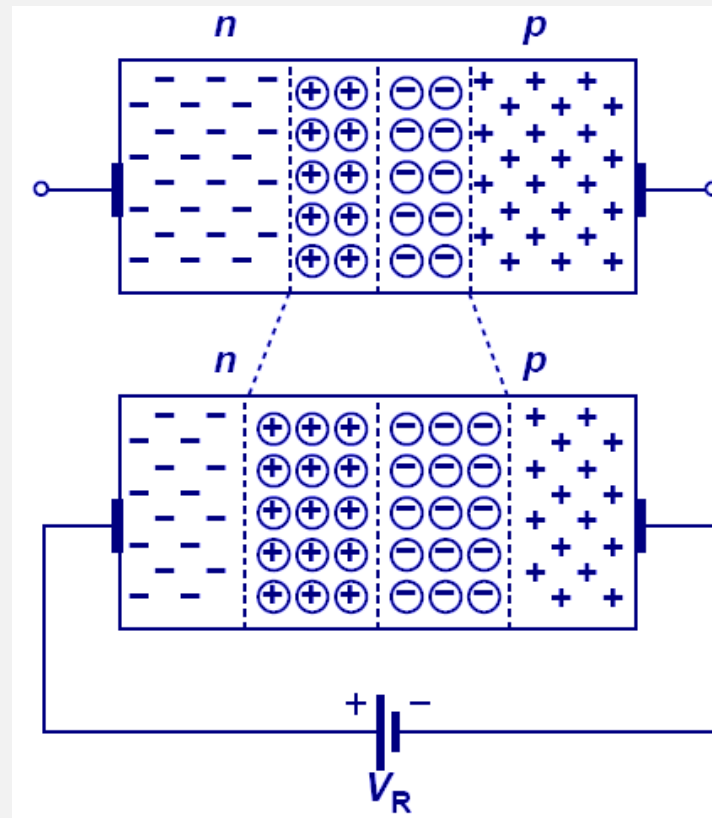
- V_0 is a **weak function** of the doping levels (N_A or N_D). How much does V_0 change if N_A or N_D is increased by one order of magnitude?

Solution:

$$\begin{aligned}\Delta V_0 &= \left(V_T \ln \frac{10 N_A N_D}{n_i^2} \right) - \left(V_T \ln \frac{N_A N_D}{n_i^2} \right) \\ &= V_T \left(\ln \frac{10 N_A N_D}{n_i^2} - \ln \frac{N_A N_D}{n_i^2} \right) \\ &= V_T \ln \left(\frac{10 N_A N_D}{n_i^2} / \frac{N_A N_D}{n_i^2} \right) \\ &= V_T \ln 10 \\ &= 60 \text{ mV}\end{aligned}$$

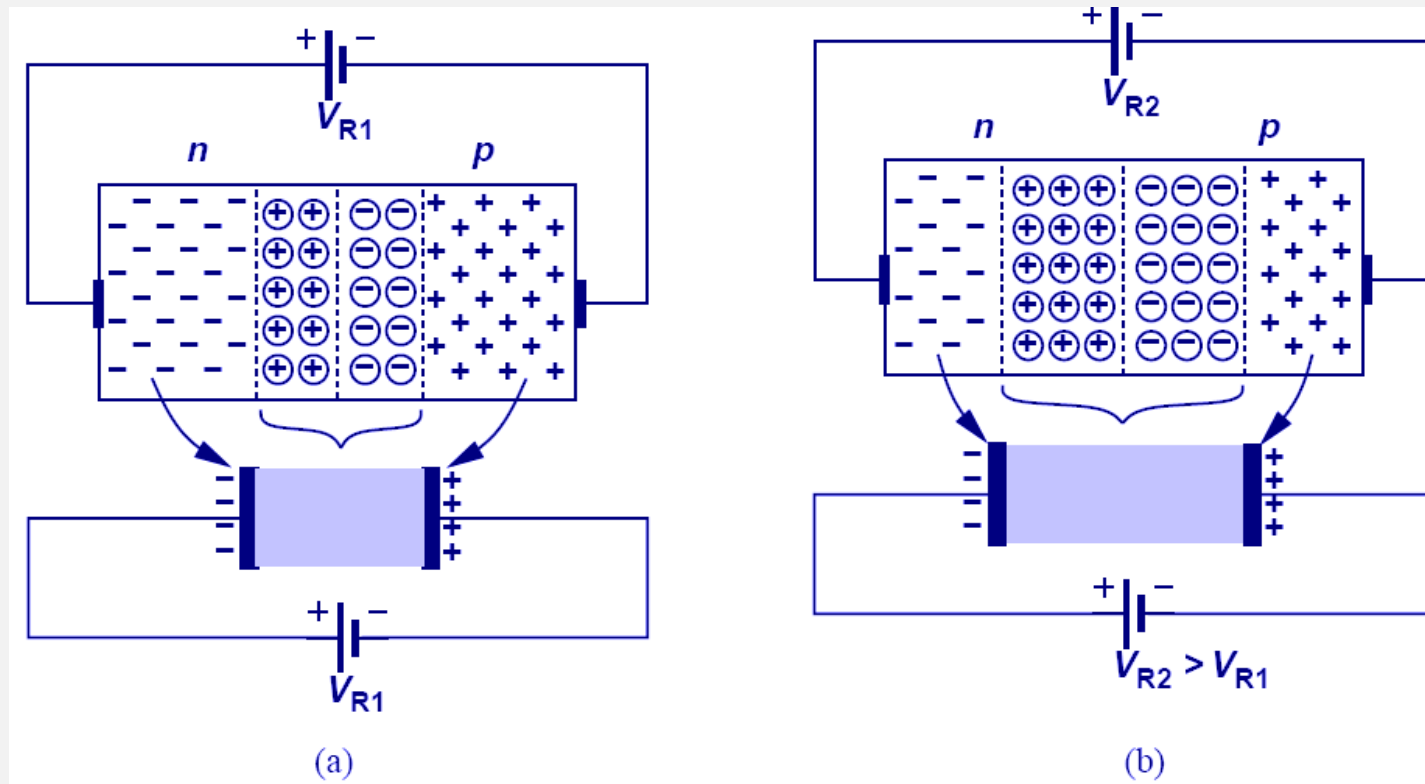
Diode in Reverse Bias

- When the N-type region of a diode is connected to a higher potential than the P-type region, the diode is under reverse bias, which results in **wider depletion region** and **larger built-in electric field** across the junction.



Reverse Biased Diode's Application: Voltage-Dependent Capacitor

- The PN junction can be viewed as a capacitor. By varying V_R , the depletion width changes, changing its capacitance value; therefore, the PN junction is actually a voltage-dependent capacitor.



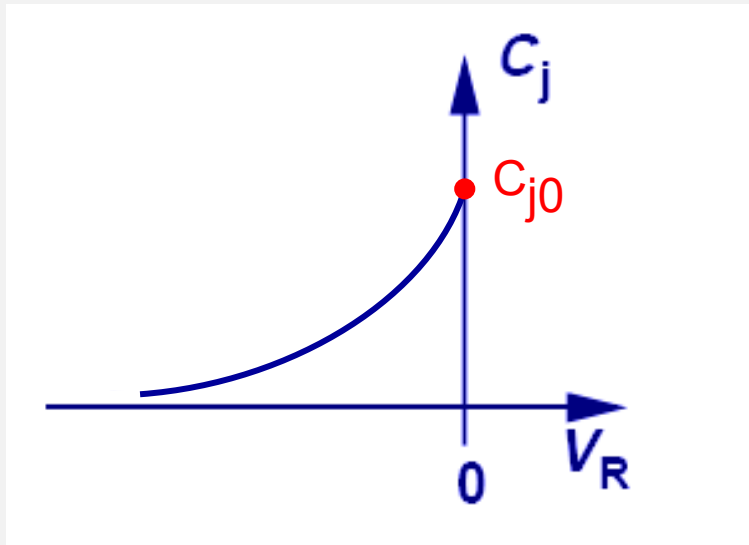
Reduction of junction capacitance with reverse bias

Voltage-Dependent Capacitance

- The equations that describe the voltage-dependent capacitance are shown below.

$$C_j = \frac{C_{j0}}{\sqrt{1 - \frac{V_R}{V_0}}}$$

where C_{j0} denotes the capacitance corresponding to zero bias ($V_R = 0$ V), and



$$C_{j0} = \sqrt{\frac{\epsilon_{si} q}{2} \frac{N_A N_D}{N_A + N_D} \frac{1}{V_0}}$$

where $\epsilon_{si} = 11.7 \times (8.85 \times 10^{-14} \text{ F/cm})$ represents the dielectric constant of silicon.

- The dielectric constant of materials is usually written in the form $\epsilon_r \epsilon_0$, where ϵ_r is the “relative” dielectric constant ϵ_0 the dielectric constant of vacuum ($8.85 \times 10^{-14} \text{ F/cm}$)

Example 2-15

- A pn junction is doped with $N_A = 2 \times 10^{16} \text{cm}^{-3}$ and $N_D = 9 \times 10^{15} \text{cm}^{-3}$. Determine the capacitance of the device with (a) $V_R = 0 \text{ V}$ and (b) $V_R = -1 \text{ V}$.

Solution:

$$V_0 = V_T \ln \frac{N_A N_D}{n_i^2} = (26 \text{mV}) \ln \frac{(2 \times 10^{16}) \times (9 \times 10^{15})}{(1.08 \times 10^{10})^2} \\ = 730 \text{ mV}$$

$$C_{j0} = \sqrt{\frac{\epsilon_{si} q}{2} \frac{N_A N_D}{N_A + N_D} \frac{1}{V_0}} \\ = \sqrt{\frac{(11.7 \times (8.85 \times 10^{-14})) \times (1.6 \times 10^{-19})}{2} \frac{(2 \times 10^{16}) \times (9 \times 10^{15})}{2 \times 10^{16} + 9 \times 10^{15}} \frac{1}{0.73}} = 2.65 \times 10^{-8} \text{ F/cm}^2 \\ = 0.265 \text{ fF}/\mu\text{m}^2 \quad (\text{fF} = 10^{-15} \text{ F})$$

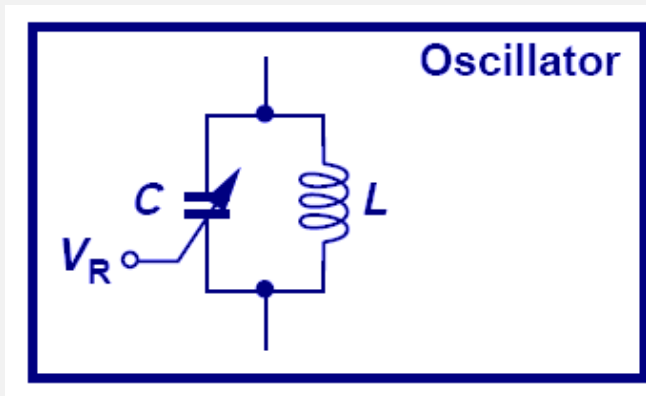
$$\text{(a) } V_R = 0, C_j = 0.265 \text{ fF}/\mu\text{m}^2, \quad \text{(b) } V_R = -1, C_j = \frac{C_{j0}}{\sqrt{1 + \frac{1}{V_0}}} = 0.172 \text{ fF}/\mu\text{m}^2$$

Example 2-16: Voltage-Controlled Oscillator

- A cellphone incorporates a 2-GHz oscillator whose frequency is defined by the resonance frequency of an LC tank. If the tank capacitance is realized as the pn junction of Example 2-15, calculate the change in the oscillation frequency while the reverse voltage goes from 0 to 2 V. Assume the circuit operates at 2 GHz at a reverse voltage of 0 V, and the junction area is $2000 \mu\text{m}^2$.

Solution:

From basic circuit theory that the tank “resonates” if the impedances of the inductor and the capacitor are equal and opposite, i.e.,



Variable capacitor used to tune an oscillator

$$X_C = -X_L, \text{ where } X_C = \frac{1}{j\omega C} \text{ and } X_L = j\omega L.$$

Thus, the resonance frequency is equal to

$$f_{res} = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

At $V_R = 0$ V, $C_j = 0.265$ fF/ μm^2 , the device capacitance equals

$$\begin{aligned} C &= C_j(V_R = 0 \text{ V}) \times \text{Area} \\ &= 0.265 \text{ fF}/\mu\text{m}^2 \times 2000 \mu\text{m}^2 \\ &= \mathbf{530 \text{ fF}} \end{aligned}$$

$$\therefore f_{res} = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \Rightarrow 2 \times 10^9 = \frac{1}{2\pi} \frac{1}{\sqrt{L \times (530 \times 10^{-15})}}$$

$$\mathbf{L = 11.9 \text{ nH}}$$

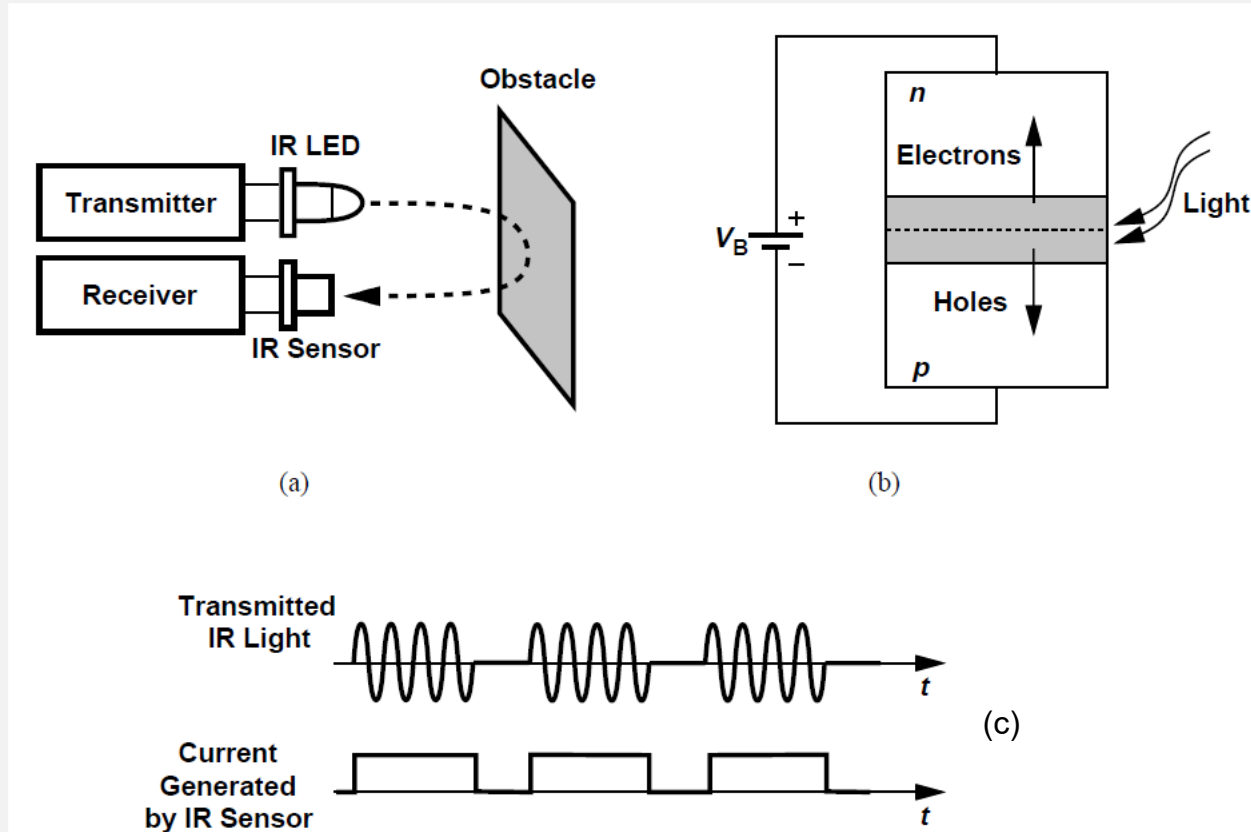
When $V_R = -2$ V,

$$\begin{aligned} C &= C_j(V_R = -2 \text{ V}) \times \text{Area} = \frac{C_{j0}}{\sqrt{1 - \frac{(-2)}{0.73}}} \times 2000 \mu\text{m}^2 \\ &= \mathbf{274 \text{ fF}} \end{aligned}$$

$$\begin{aligned} f_{res}(V_R = -2 \text{ V}) &= \frac{1}{2\pi} \frac{1}{\sqrt{(11.9 \times 10^{-9}) \times (274 \times 10^{-15})}} \\ &= \mathbf{2.78 \text{ GHz}} \end{aligned}$$

Robotics Application: Infrared Transceivers

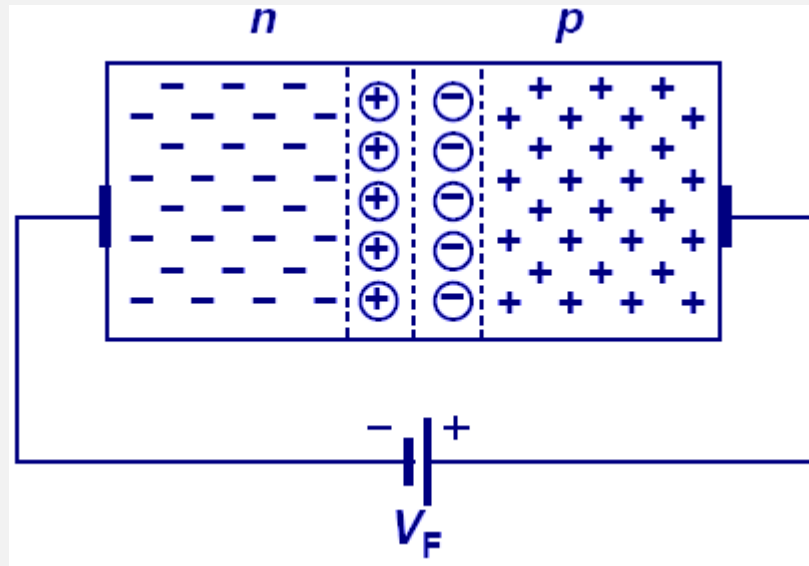
- Some robots employ infrared (IR) transceivers to detect obstacles, control their distance with respect to other robots, or transmit and receive data.



(a) IR transceiver system, (b) operation of IR sensor, and (c) pulsed light and resulting sensor current.

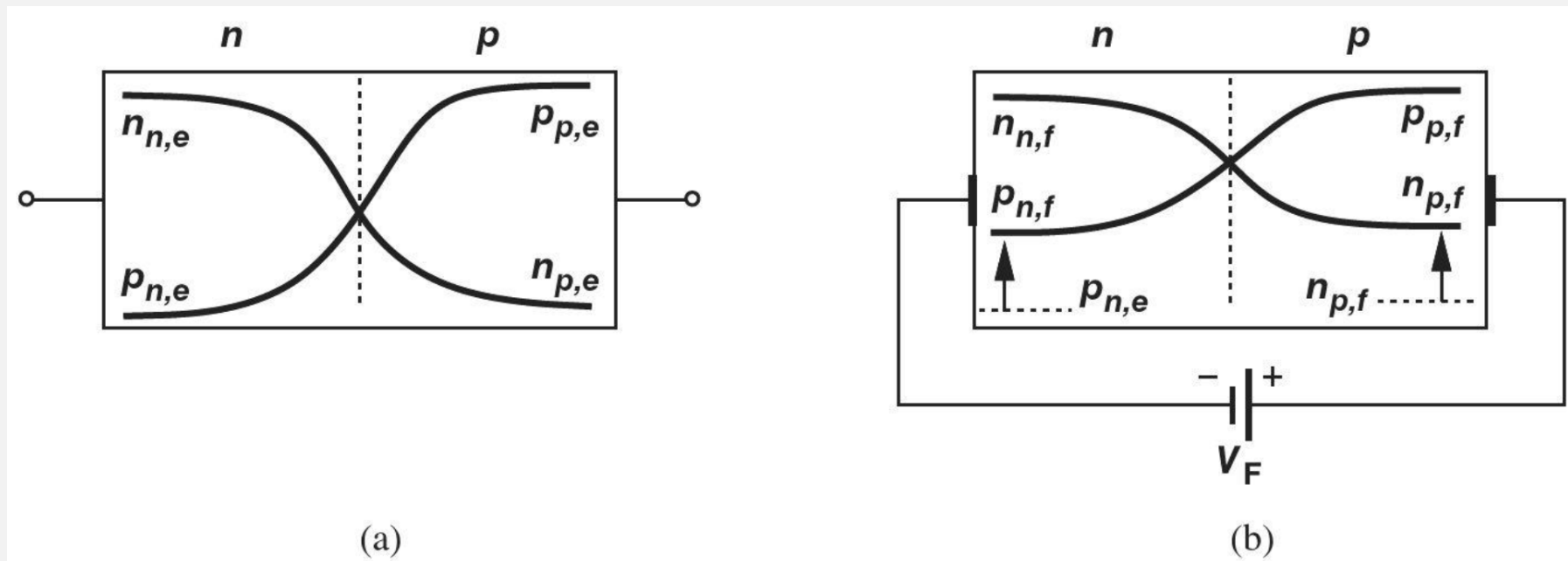
Diode in Forward Bias

- When the N-type region of a diode is at a lower potential than the P-type region, the diode is in forward bias.
- The depletion width is shortened and the built-in electric field decreased.



Minority Carriers Increase in Forward Bias

- Under forward bias, **minority carriers in each region increase** due to the **lowering of built-in field/potential**. Therefore, diffusion currents increase to supply these minority carriers.



Carrier profiles (a) in equilibrium and (b) under forward bias

$$\because V_0 = V_T \ln \frac{p_p}{p_n} \Rightarrow \frac{p_p}{p_n} = e^{\frac{V_0}{V_T}}$$

In equilibrium condition,

$$p_{n,e} = p_{p,e} e^{-\frac{V_0}{V_T}}$$

In forward bias,

$$p_{n,f} = p_{p,f} e^{-\left(\frac{V_0 - V_F}{V_T}\right)}$$

$$\because p_{p,e} \approx p_{p,f} \approx N_A$$

$$p_{n,f} = e^{V_F/V_T} \times p_{n,e} \gg p_{n,e}$$

On the n side, the hole concentration increases as

$$\begin{aligned} \Delta p_n &= p_{n,f} - p_{n,e} = (e^{V_F/V_T} - 1) p_{n,e} \\ &\approx (e^{V_F/V_T} - 1) N_A e^{-V_0/V_T} \end{aligned}$$

Similarly, on the p side, the electron concentration increases as

$$\begin{aligned}\Delta n_p &= (e^{V_F/V_T} - 1)n_{p,e} \\ &\approx (e^{V_F/V_T} - 1)N_D e^{-V_0/V_T}\end{aligned}$$

The total diffusion current increases by

$$I_{tot} \propto \Delta p_n + \Delta n_p$$

It can be proved that

$$I_{tot} = I_s(e^{V_F/V_T} - 1)$$

where I_s called the “reverse saturation current”. And I_s is given by

$$I_s = Aqn_i^2\left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p}\right).$$

where A is the cross section area of the device, and L_n and L_p are electron and hole “diffusion lengths,” respectively.

Example 2-17

- Determine I_s for the junction of Example 2-13 at $T = 300$ K if $A = 100 \mu\text{m}^2$, $L_n = 20 \mu\text{m}$, and $L_p = 20 \mu\text{m}$.

Solution:

For intrinsic silicon,

$$n_i^2 = 1.08 \times 10^{10} \text{ electrons/cm}^3$$

$$D_n = 34 \text{ cm}^2/\text{s}, D_p = 12 \text{ cm}^2/\text{s}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

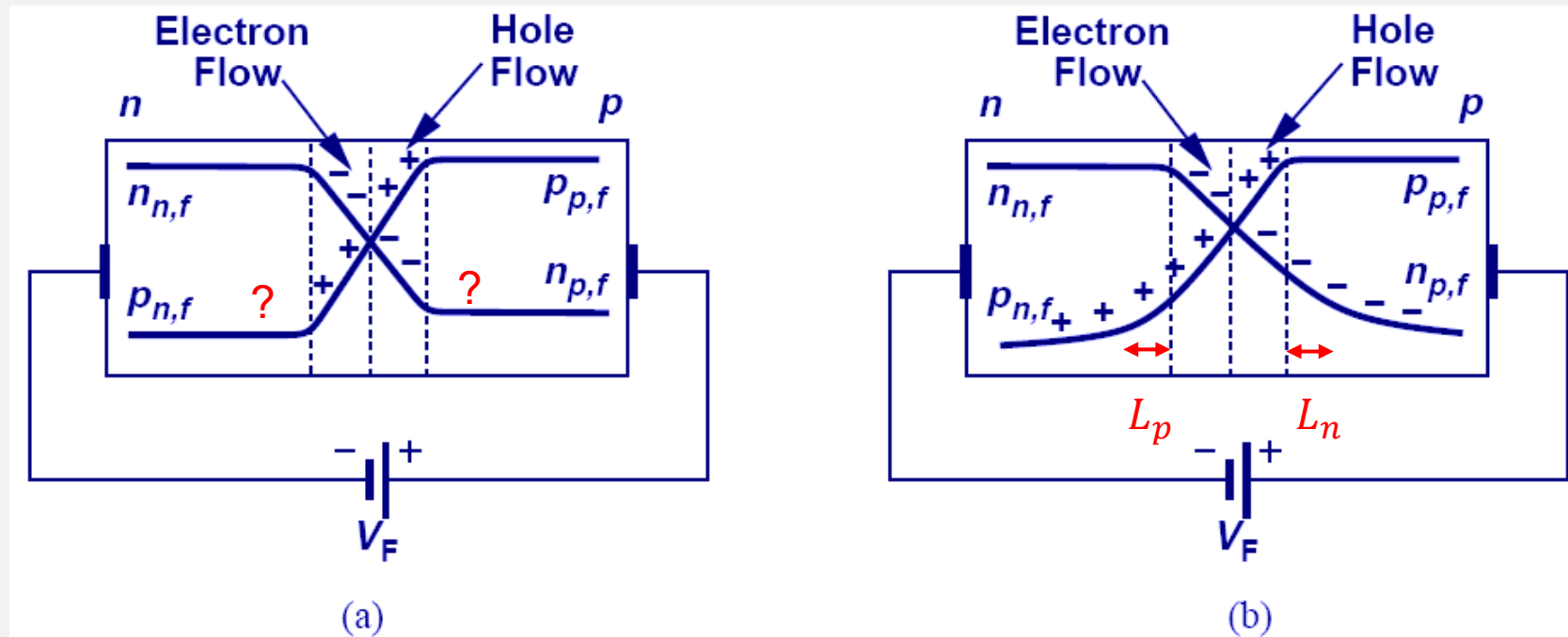
From Example 2-13, $N_A = 2 \times 10^{16} \text{ cm}^{-3}$, $N_D = 4 \times 10^{16} \text{ cm}^{-3}$

$$I_s = Aqn_i^2 \left(\frac{D_n}{N_AL_n} + \frac{D_p}{N_DL_p} \right) = 1.77 \times 10^{-17} \text{ A}$$

- ✓ Since I_s is extremely small, the exponential term $(e^{V_F/V_T} - 1)$ must be very large values so as to yield a useful amount (e.g., 1 mA) for I_{tot} .

Minority Carriers Increase in Forward Bias

- Minority carrier concentration profile **should not be constant** along the x-axis; otherwise, there is **no concentration gradient** and **no diffusion** current.

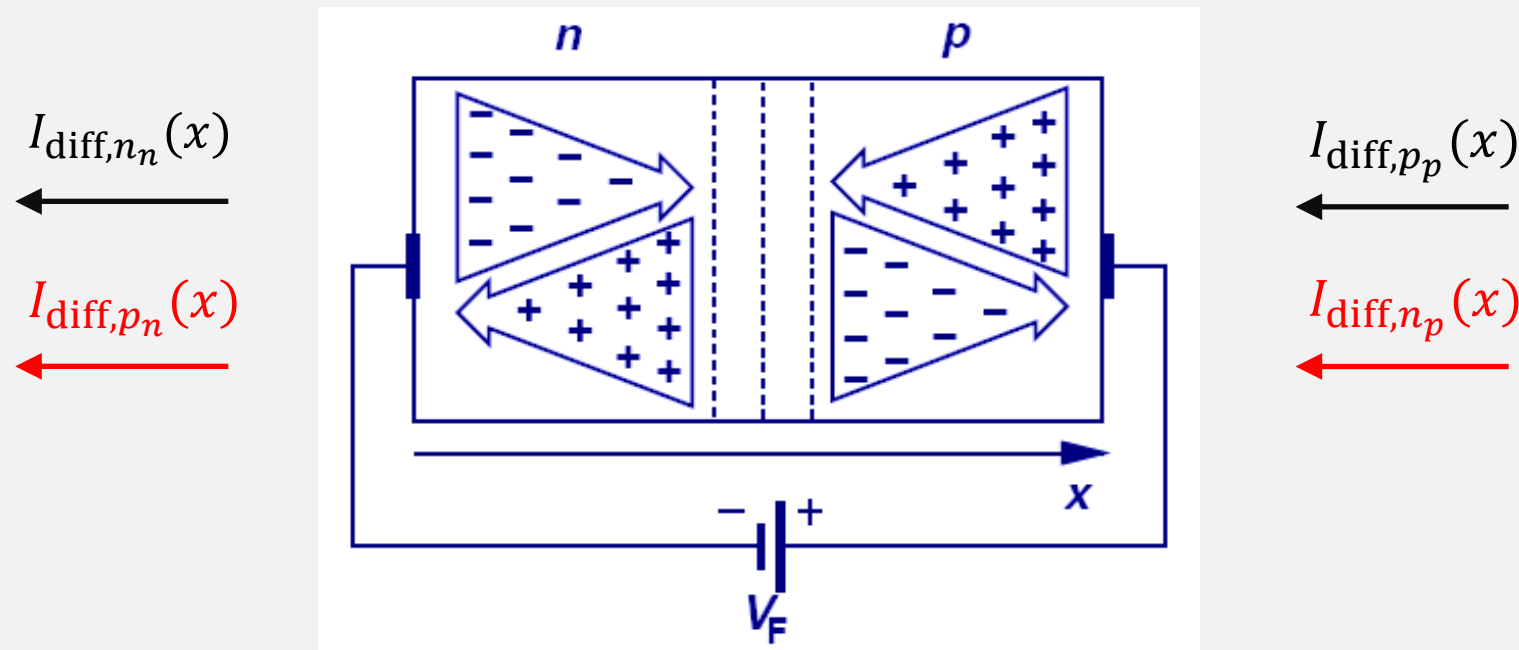


(a) Constant and (b) **variable** majority carrier profiles outside the depletion region.

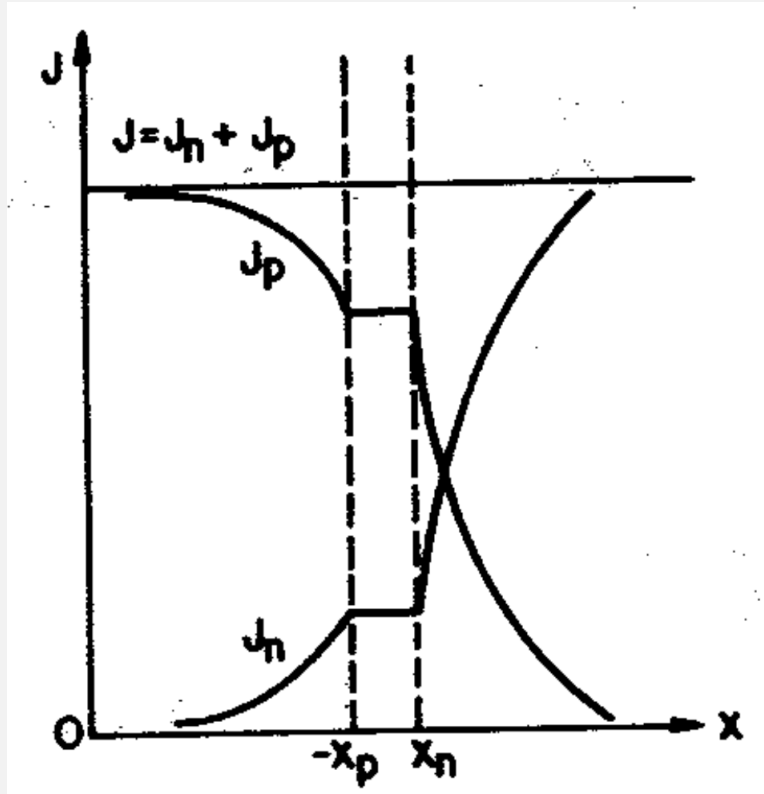
Sum of Diffusion Currents is a Constant in Forward Bias

- In forward bias, minority carriers diffuse forward and gradually decrease due to recombination with majority carriers. Majority carriers decrease similarly. Both currents sum to a constant value.

$$I_{\text{diff},n_n}(x) + I_{\text{diff},p_n}(x) = \text{constant} = I_{\text{diff},p_p}(x) + I_{\text{diff},n_p}(x)$$



Profile of Diffusion Current Density in Forward Bias



$$J = J_n(x) + J_p(x) = \text{constant}$$

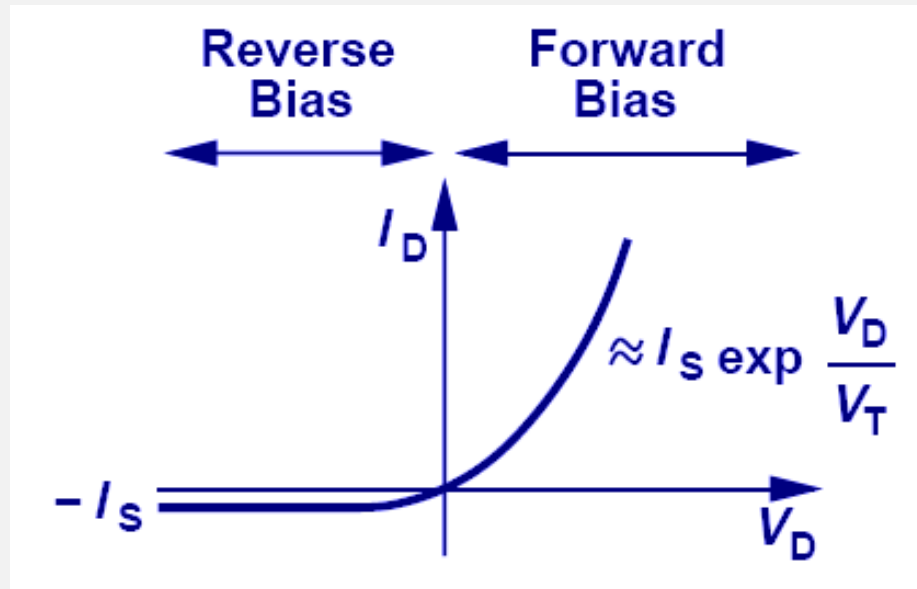
$J_{diff,n}(x)$
←

$J_{diff,p}(x)$
←

Source: S. M. Sze, Physics of Semiconductor Devices, 2nd ed., 1981.

I/V Characteristic of PN Junction

- The current and voltage relationship of a PN junction is exponential in forward bias region, and relatively constant in reverse bias region.



$$I_D = I_S \left(\exp \frac{V_D}{V_T} - 1 \right) \\ \approx I_S \exp \frac{V_D}{V_T}$$

Example 2-18 Parallel PN Junctions

- Each junction in Fig. 2.33 employs the doping levels described in Example 2-13. Determine the forward bias current of the composite device for $V_D = 300$ mV and 800 mV at $T = 300$ K.

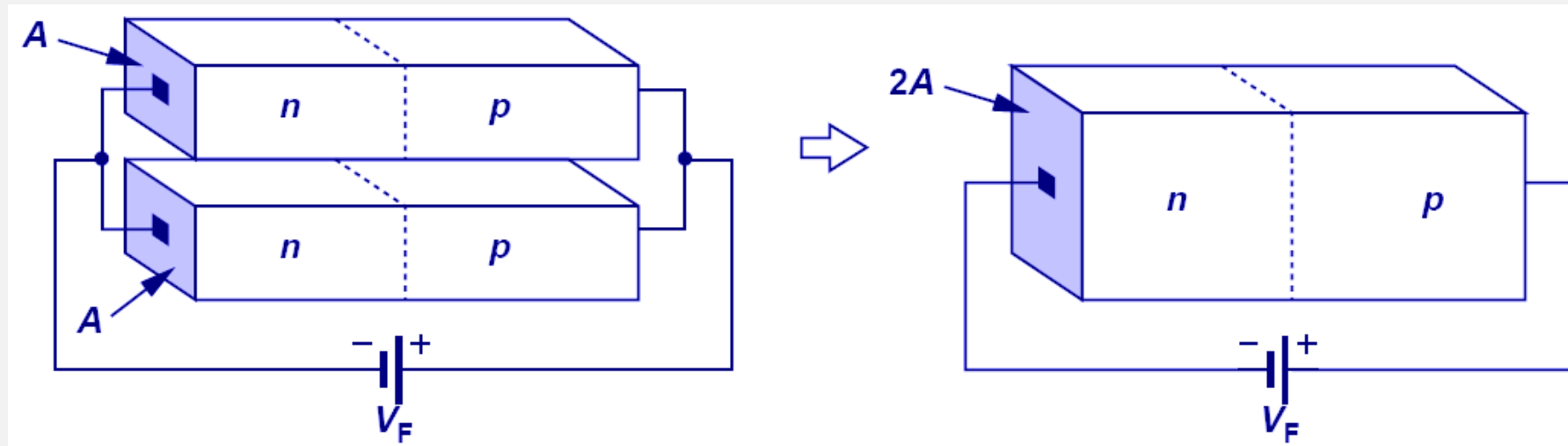


Figure 2.33 Equivalence of parallel devices to a larger device.

Solution:

From Example 2-17, $I_S = 1.77 \times 10^{-17} \text{ A}$, the total current is equal to

$$\begin{aligned} I_{D,tot}(V_D = 300 \text{ mV}) &= 2I_S \left(\exp \frac{V_D}{V_T} - 1 \right) \\ &= 2 \times (1.77 \times 10^{-17}) \left(\exp \frac{300 \text{ mV}}{26 \text{ mV}} - 1 \right) \\ &= 3.63 \text{ pA} \end{aligned}$$

$$\begin{aligned} I_{D,tot}(V_D = 800 \text{ mV}) &= 2 \times (1.77 \times 10^{-17}) \left(\exp \frac{800 \text{ mV}}{26 \text{ mV}} - 1 \right) \\ &= 816 \text{ } \mu\text{A} \end{aligned}$$

Example 2-19

- A diode operates in the forward bias region with a typical current level [i.e., $I_D \approx I_S \exp(V_D/V_T)$]. Suppose we wish to increase the current by a factor of 10. How much change in V_D is required?

Solution:

$$\because I_D = I_S \exp(V_D/V_T)$$

$$V_D = V_T \ln(I_D/I_S)$$

$$\Delta V_D = V_{D1} - V_D = V_T \ln 10$$

$$= (26 \text{ mV}) \ln 10$$

$$= 59.9 \text{ mV}$$

$$\text{For } I_1 = 10I_D, V_{D1} = V_T \ln(I_1/I_S)$$

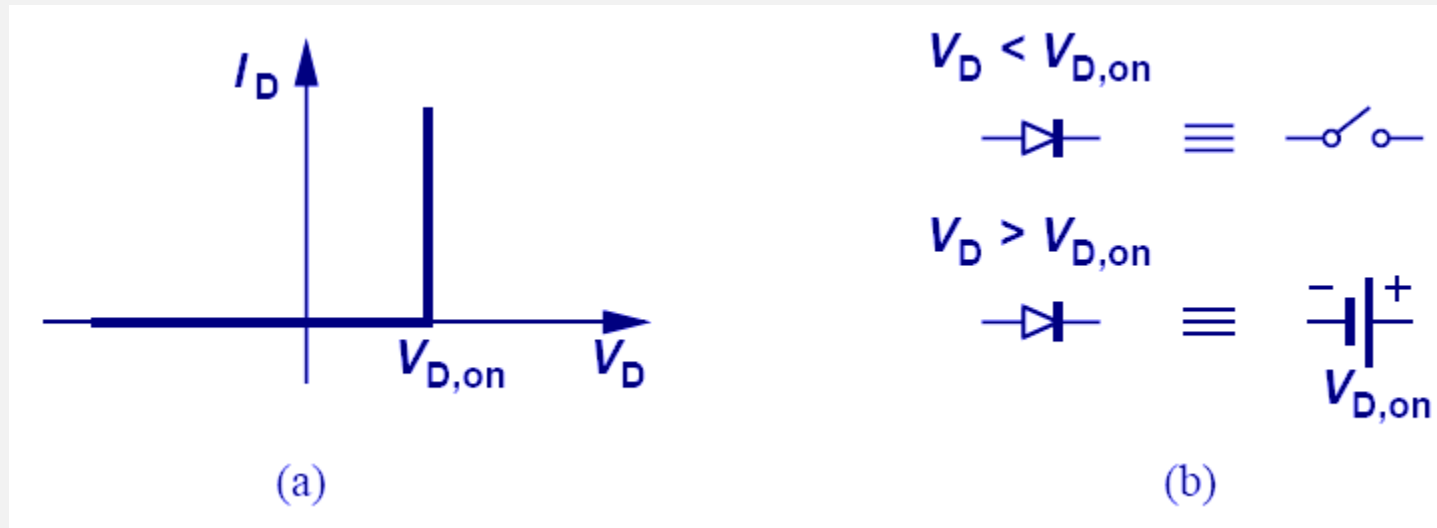
$$V_{D1} = V_T \ln(10I_D/I_S)$$

$$= V_T \ln 10 + V_T \ln(I_D/I_S)$$

$$= V_T \ln 10 + V_D$$

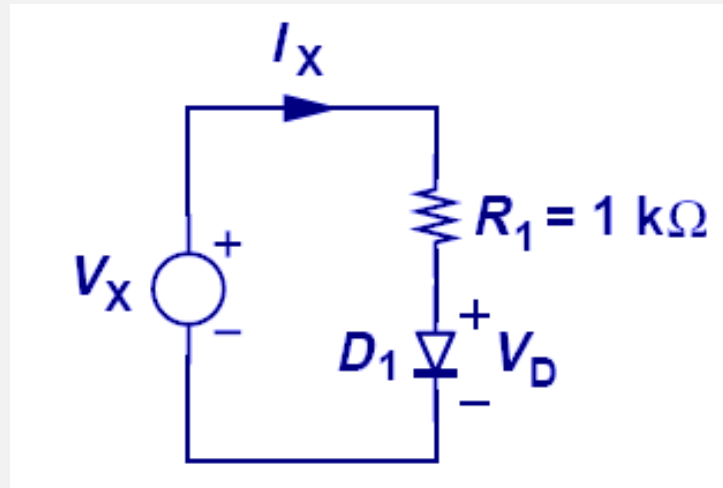
Constant-Voltage Diode Model

- Diode operates as an open circuit if $V_D < V_{D,on}$ and a constant voltage source of $V_{D,on}$ if V_D tends to exceed $V_{D,on}$.



Example 2-21: Nonlinear vs. Linear Diode Model

- Consider the following circuit. Calculate I_X for $V_X = 3\text{ V}$ and $V_X = 1\text{ V}$ using (a) an exponential model with $I_S = 10^{-16}\text{ A}$ and (b) a constant-voltage model with $V_{D,on} = 800\text{ mV}$.



Solution:

(a) For exponential model:

By KVL, $V_X = I_X R_1 + V_D$.

$$V_X = I_X R_1 + V_T \ln \frac{I_X}{I_S}$$

This equation is nonlinear and must be solved iteratively, rewrite I_X as

$$I_X = \frac{V_X - V_D}{R_1}$$

Guess $V_D = 750 \text{ mV}$, $I_X = \frac{3 \text{ V} - 0.75 \text{ V}}{1 \text{ k}\Omega} = 2.25 \text{ mA}$

$$V_D = V_T \ln \frac{I_X}{I_S} = 26 \text{ mV} \ln \frac{2.25 \text{ mA}}{10^{-16} \text{ A}} = 779 \text{ mV}$$

$$I_X = \frac{V_X - V_D}{R_1} = \frac{3 \text{ V} - 0.779 \text{ V}}{1 \text{ k}\Omega} = 2.201 \text{ mA}$$

Similarly, for $V_X = 1 \text{ V}$,

$$I_X = \frac{1 \text{ V} - 0.75 \text{ V}}{1 \text{ k}\Omega} = 0.25 \text{ mA}$$

$$V_D = 26 \text{ mV} \ln \frac{0.25 \text{ mA}}{10^{-16} \text{ A}} = 742 \text{ mV}$$

$$I_X = \frac{1 \text{ V} - 0.742 \text{ V}}{1 \text{ k}\Omega} = 0.258 \text{ mA}$$

(b) For constant-voltage model:

By KVL, $V_X = I_X R_1 + V_D$.

$$I_X = \frac{V_X - V_D}{R_1}$$

$$I_X = \frac{3\text{ V} - 0.8\text{ V}}{1\text{ k}\Omega} = 2.2\text{ mA}$$

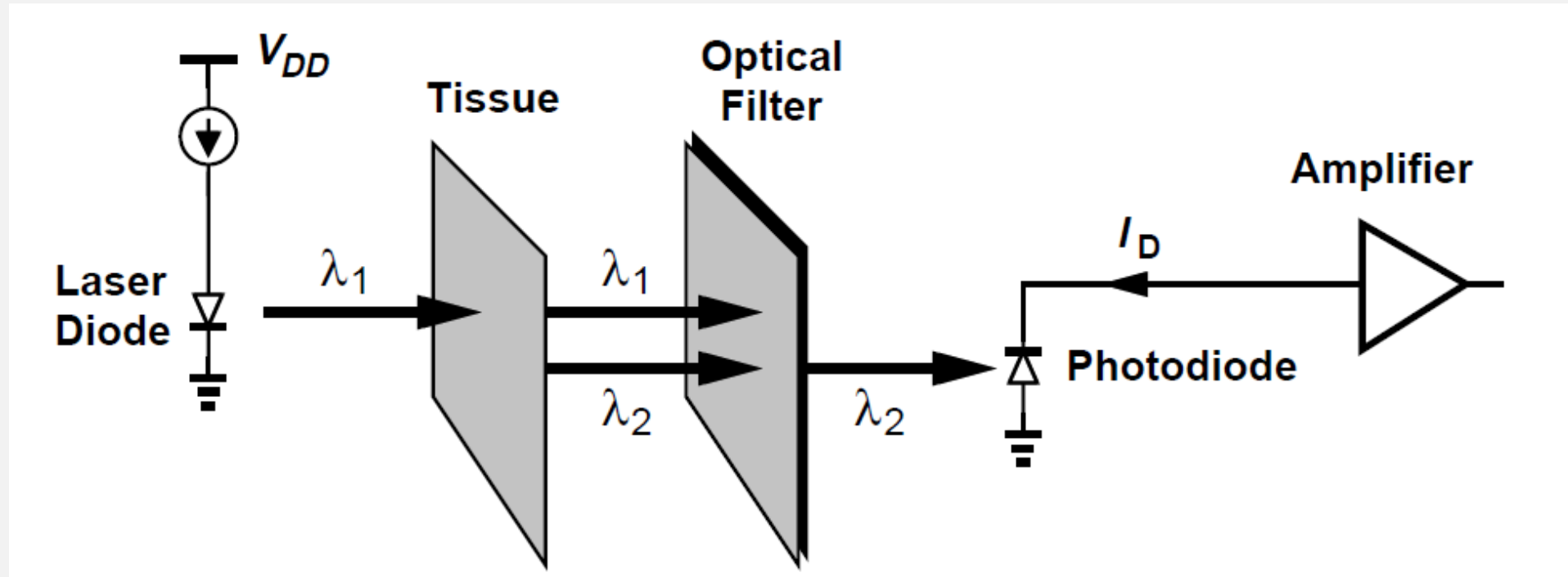
Similarly, for $V_X = 1\text{ V}$,

$$I_X = \frac{1\text{ V} - 0.8\text{ V}}{1\text{ k}\Omega} = 0.2\text{ mA}$$

- ✓ This example shows the **simplicity** provided by a constant-voltage model over an exponential model.
- ✓ For an exponential model, **iterative method** is needed to solve for current, whereas constant-voltage model requires only linear equations.

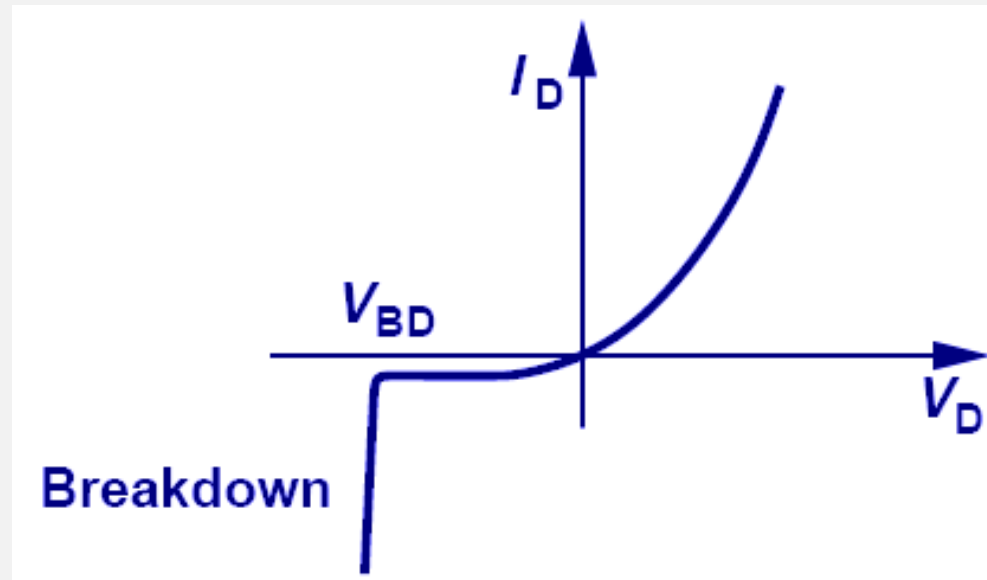
Bioengineering Application: Cancer Detection

- Laser diode generates light with wavelength λ_1 . If the tissue has cancerous cells and contains “fluorophore,” then it generates a new wavelength, λ_2 .



Reverse Breakdown

- When a large reverse bias voltage is applied, breakdown occurs and an enormous current flows through the diode.



Zener vs. Avalanche Breakdown

- The breakdown phenomenon in pn junctions occurs by one of two possible mechanisms: “Zener effect” and “avalanche effect.”
- Zener breakdown is a result of the **large electric field** inside the depletion region that breaks electrons or holes off their covalent bonds.
- Avalanche breakdown is a result of electrons or holes **colliding with the fixed ions** inside the depletion region.

